

# HAMZA BUTT

Senior Backend / Platform Engineer

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## PROFESSIONAL SUMMARY

Backend / Platform Engineer with 4+ years of experience building scalable Node.js systems across IoT, SaaS, and compliance-driven platforms. Specialized in high-throughput data pipelines, multi-tenant architectures, and privacy frameworks (GDPR/CCPA). Proven ability to design and deliver production systems handling 10K+ daily transactions, optimize performance, and lead backend development in fast-paced environments.

## CORE TECHNICAL SKILLS

**Backend:** Node.js, Express.js, Fastify, TypeScript, REST APIs, JWT

**Databases:** PostgreSQL, MySQL, MongoDB, ClickHouse, Prisma, Sequelize

**Cloud & DevOps:** AWS (S3, EC2), Docker, Podman, GitHub Actions, Self-hosted GitHub Runners, CI/CD, Linux

**Frontend:** Vue.js, Nuxt.js, Tailwind CSS

**Other:** MQTT, WebSockets, Kafka, Multi-tenant Architecture, System Design, Event-driven Systems

## KEY ACHIEVEMENTS

- Reduced deployment time by 87% (2 hours to 15 minutes) through CI/CD automation
- Designed IoT pipeline processing 10,000+ events/minute for predictive maintenance
- Built backend systems handling 50,000+ daily transactions
- Architected GDPR/CCPA compliance framework across multiple product lines
- Led and mentored junior engineers to production readiness

## PROFESSIONAL EXPERIENCE

### Senior Software Engineer

April 2025 – Present

*OnStak Inc.*

#### Video AI Platform (Next-Gen Revamp)

*Node.js, Express, PostgreSQL, Kafka, ClickHouse*

- Led backend development of next-generation Video AI platform with advanced AI-driven capabilities
- Designed and implemented high-throughput, event-driven ingestion pipeline using Kafka and ClickHouse for real-time video analytics
- Built intelligent investigation engine enabling automated case analysis from video events and behavioural patterns
- Integrated AI-powered chatbot assistant supporting natural language querying and investigation workflows
- Migrated and optimized core services to PostgreSQL, improving data consistency and query performance
- Architected scalable, distributed backend services using Node.js (Express) for high-throughput video event processing

#### Regulatory Compliance & Privacy Framework

- Architected GDPR/CCPA compliance framework across 3 product lines, implementing encryption, automated data retention, and right-to-deletion workflows
- Designed audit trail system tracking 100% of data access events, reducing audit preparation time by 60%
- Led cross-functional implementation of cookie consent and privacy systems

#### AI Predictive Maintenance Platform

*Node.js, PostgreSQL, AWS S3, Docker*

- Designed and scaled IoT data pipeline processing 10,000+ events/minute for real-time failure prediction
- Architected secure multi-tenant system with row-level isolation and RBAC for enterprise clients
- Automated CI/CD pipelines using GitHub Actions, reducing deployment time from 2 hours to 15 minutes

#### Player Analytics Platform (Cricket Board)

*Node.js, Fastify, MySQL, Prisma, MQTT*

- Served as technical lead for backend team of 3 engineers
- Implemented real-time notifications using MQTT for 500+ concurrent users
- Built containerized CI/CD pipeline using Podman
- Mentored junior developers to production readiness

#### AI Chatbot Platform Migration

*Supabase, Vercel, GitHub*

- Executed zero-downtime database migration preserving authentication and user sessions
- Automated deployment workflows, reducing manual release effort by 80%

## Junior Software Engineer

April 2022 – April 2025

*OnStak Inc.*

### Drive-Thru Analytics Platform

*Node.js, Express, MySQL, Sequelize*

- Developed backend handling 50,000+ daily transactions across 30+ POS integrations
- Built real-time analytics and reporting system for operational insights

### Video Analytics Platform

*Node.js, Express, MySQL, AWS S3*

- Integrated multiple computer vision models into production backend
- Designed tenant-based authorization system supporting 15+ enterprise clients

## Associate Software Engineer

July 2021 – April 2022

*Wi-Metrix*

### Production Line Management System

*Node.js, Express, SQL Server, Sequelize*

- Improved query performance by 40% through indexing and query optimization
- Implemented caching layer reducing database load by 25%
- Delivered APIs supporting multiple frontend and mobile applications

## EDUCATION

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### Bachelor of Science in Computer Science (BSCS)

Forman Christian College, Lahore | 2016 – 2020

- Final Year Project: Social Media Predictive Analytics System (ML-based, 85% accuracy)
- Relevant Coursework: Data Structures, Algorithms, Database Systems, Software Engineering

## CERTIFICATIONS

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- Introduction to Data Analytics for Business — University of Colorado Boulder (2024)

## ADDITIONAL INFORMATION

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**Languages:** English (Professional), Urdu (Native)

**Interests:** Technical content creation, photography, competitive gaming